Project Report

OPEN-SOURCE SOFTWARE

(LINUX ADMINISTRATION AUDIT)

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Course: Open-Source Software

Software: Git

INTRODUCTION

Today, the days of Individual Programmers developing software on their own have long since passed. Instead, software development today involves Collaboration between Developers from all over the world who work together for a single purpose: To Create Software. This change has been made possible through, largely due to, an increase in the availability of Open-Source Software.

Open-Source Software allows anyone to access, Change, and Share the source code of the software they are using. This allows for Creativity, Education, and Collaboration. The core aspect of this new paradigm is Git.

Git is a Version Control System that provides Developers with a means to control and track the changes made to their code. Because Git is a distributed version control system, it allows Multiple Developers to collaborate on a single project without interfering with the changes each other has made. Also, unlike traditional version control systems, Git provides a complete copy of the entire Repository to all Users, so the same files can be modified by Multiple Users without Issue.

This document will provide as in-depth of a discussion about Git as possible. I will examine not only the Features and Functionalities of Git but also why Git was Developed in the first place and how Git is a key contributor to the Open-Sourced Software Movement.

PART A

ORIGIN AND PHILOSOPHY

A1) The Problem Git Was Created to Solve

Before the invention of Git, various tools such as CVS and Subversion were used for version control. These tools relied on the central server for storing all the project information. Even though this system was effective for small teams, various problems were associated with it.

One such problem was the dependency on the server. If server crashed, the developers were not able to continue their work. This led to the creation of a single point of failure. Another drawback was the slow process of branching and merging, which made developers feel as though they were restricted.

Linus Torvalds developer of the Linux, encountered a major issue when proprietary system known as BitKeeper was no longer for free use. This was a serious issue as the Linux project required an efficient type of control system.

Instead of depending on another existing system, Linus decided to develop another system from scratch. This led to the invention of the version control system known as Git in 2005. The main goal was on performance, reliability and simplicity.

Git brought about the idea of distributed version control. This means that every developer had a full copy of the project, including the version history. This allowed them to work independently and then merge the code.

Another important aspect of Git was the integrity of the data. It used cryptographic hashes to track all the data. This ensured the security and integrity of the data.

Thus, Git overcame all the limitations of earlier systems and offered an efficient means of managing projects.

A2) The License — What It Actually Says

The project is released under GNU General Public License version 2 which is also known as the GPL version 2. This ensures that software is open source and free.

The four main freedoms provided by GPL are:

- The freedom to run the software for any purpose.
- The freedom to study the source code of the software.
- The freedom to modify the source code of the software.
- The freedom to distribute the original and/or modified version(s) of the software.

One of the important characteristics of the GPL is that the license is considered to be a copyleft license. The definition of the copyleft license is as follows: “A copyleft license is a license for software that requires any derivative work created from the original software and distributed by a person to include a certain amount of source code for distribution.” The purpose of the GPL is to ensure that the contributions made by the users are made available to the entire user community.

One of the important differences between the GPL and the free software license is that the derivative work of the software distributed by the person is better protected by the GPL than the free software license, i.e., the MIT License and the Apache License. The important difference between the GPL and the free software license is that the GPL is focused on maintaining and preserving the freedoms of the user.

The definition of the free software is as follows: “Free software” means that the user is free to use the software by exercising the four main freedoms provided by the GPL.

A3) The Ethics of Open Source

However, open source is not only concerned with technology, but also with values such as sharing, transparency, and collaboration.

There are a number of benefits to open source software. Firstly, open source encourages innovation because developers can improve on what has already been done. Secondly, open source eliminates the need to be dependent on a company and also provides a way to customize software to meet one's needs.

However, I also identified some challenges to open source software. For example, some companies have been using open source software to create a successful product and then not giving back to the community. This, although legal, is not ethical.

In my opinion, open source software should be supported at both individual and organizational levels. Therefore, if a company is benefiting from open source, they should also contribute to the community, either by improving the software or giving back to the community.

The phrase "standing on the shoulders of giants" was coined to mean exactly what open source is. Developers improve on what has already been done to create a better solution, thus allowing technology to advance at a faster rate.

PART B

LINUX FOOTPRINT

It operates with maximum efficiency in Linux-based systems. It is easy to install on Linux-based systems, as the package managers such as ‘apt’ support the installation process.

Once the installation process is complete, the user can use the git program. It does not require program to be run as a separate process, which is an advantage.

The location where the ‘git’ program is installed is ‘/usr/bin/git.’ The configuration files are also located in different places depending on the purpose for which the program is used. The system level configuration located at ‘/etc/gitconfig,’ while the user level configuration is located in the home directory with the name ‘~/.gitconfig.’

It operates as the current user and there is no need to provide any special privileges.

The updates are provided by package manager. The open-source community keeps on updating the program to remove bugs from the system.

PART C

THE FOSS ECOSYSTEM

Git plays a very important role in the open-source world. It is used in almost every project. It is also used for collaboration on a global scale.

There are many other tools, such as GitHub, GitLab, and Bitbucket, which operate on Git. They offer many other features as well.

Git is based on standard libraries and is implemented in the C language.

Git has a very large community. Many people around the world contribute to Git. They contribute by filing issues, making suggestions, and even making changes to the code.

Git has inspired many other tools and technologies. It is now the standard way to do distributed version control.

PART D

OPEN-SOURCE VS PROPRIETARY

While comparing Git and proprietary source control systems, some differences can be noted.

One of the main differences is that Git is totally free to use, whereas a license is required when using a proprietary source control system.

Another difference is transparency. Git is open source so anyone can see and understand how it is written. However, in a proprietary source control system, you cannot see how it is written.

Another difference is flexibility. Git can be customized as per requirements, whereas in a proprietary source control system, you may face restrictions.

But in a source control system like Git, you may not get support, whereas in a proprietary source control system, support is available.

Overall, I think Git is better because it gives more freedom and flexibility.

SHELL SCRIPT SECTION

```
MINGW64:/c/Users/vivan/OneDrive/Documents/Projects/oss-audit-project

vivan@VIVAN MINGW64 ~
$ cd /c/Users/vivan/OneDrive/Documents/Projects/oss-audit-project

vivan@VIVAN MINGW64 ~/OneDrive/Documents/Projects/oss-audit-project
$ ls
1.txt          README.md                       script2_package_check.sh*
2.txt          manifesto__2026-03-23.txt       script3_disk_audit.sh*
3.txt          manifesto_vivan_2026-03-23.txt  script4_log_analyzer.sh*
Footprint.txt  script1_system_info.sh*         script5_manifesto.sh*

vivan@VIVAN MINGW64 ~/OneDrive/Documents/Projects/oss-audit-project
$ |
```

Script 1

```
vivan@VIVAN MINGW64 ~/OneDrive/Documents/Projects/oss-audit-project
$ chmod +x *.sh

vivan@VIVAN MINGW64 ~/OneDrive/Documents/Projects/oss-audit-project
$ ./script1_system_info.sh
---- Linux System Identity Report ----
Kernel version: 3.6.6-1cdd4371.x86_64
Distro: Not available (running on non-native Linux environment)
User: vivan
Home Directory: /c/Users/vivan

System Uptime: Not available in this environment
Current Date & Time: Thu Mar 26 20:23:31 IST 2026

Note: Linux is open-source and generally distributed under the GPL license.

vivan@VIVAN MINGW64 ~/OneDrive/Documents/Projects/oss-audit-project
$ |
```

This script displays system information such as kernel version, user, uptime, and date. It uses variables and command substitution.

Script 2

```
vivan@VIVAN MINGW64 ~/OneDrive/Documents/Projects/oss-audit-project
$ ./script2_package_check.sh
Inspecting package: git
Git is the selected software for this audit.
Version info (fallback):
git version 2.53.0.windows.2
License: GPL-2.0

vivan@VIVAN MINGW64 ~/OneDrive/Documents/Projects/oss-audit-project
$ ./script2_package_check.sh git
Inspecting package: git
Git is the selected software for this audit.
Version info (fallback):
git version 2.53.0.windows.2
License: GPL-2.0
```

This script checks whether Git is installed and displays its details. It uses conditional statements and package manager commands.

Script 3

```
vivan@VIVAN MINGW64 ~/OneDrive/Documents/Projects/oss-audit-project
$ ./script3_disk_audit.sh
Starting disk and permission audit...
-> Directory: /etc
   Size: 1.6M
   Perms: drwxr-xr-x - Owner: vivan
----------------------------------------
Directory /var/log not found
----------------------------------------
Directory /home not found
----------------------------------------
-> Directory: /usr/bin
   Size: 91M
   Perms: drwxr-xr-x - Owner: vivan
----------------------------------------
-> Directory: /tmp
   Size: 1.5G
   Perms: drwxr-xr-x - Owner: vivan
----------------------------------------
Checking for Git config...
Found global git config at /etc/gitconfig

vivan@VIVAN MINGW64 ~/OneDrive/Documents/Projects/oss-audit-project
$
```

This script audits directories and Git configuration files. It uses loops and permission checks.

Script 4

```
vivan@VIVAN MINGW64 ~/OneDrive/Documents/Projects/oss-audit-project
$ ./script4_log_analyzer.sh script1_system_info.sh
Searching for 'error' inside script1_system_info.sh...
-----------------------------------------------------
Total matches: 0
No matches found.
-----------------------------------------------------
Size: 4.0K Perms: drwxrwxr-x - Owner: vivan
Log file size: 4.0K
-----------------------------------------------------

vivan@VIVAN MINGW64 ~/OneDrive/Documents/Projects/oss-audit-project
$ |
```

This script reads a log file and counts occurrences of a keyword. It uses a while loop and string matching.

Script 5

```
vivan@VIVAN MINGW64 ~/OneDrive/Documents/Projects/oss-audit-project
$ ./script5_manifesto.sh
--- Open Source Manifesto Generator ---
[Interactive Input Section]
All user responses are preserved
...
Generating manifesto...
Saved to manifesto_vivan_2026-03-26.txt
[Manifesto content begins]
My Open Source Manifesto
Written by Vivan on 2026-03-26

The tool I use the most is Git.
For me, freedom in open source means Independence.
In the future, I would like to build AI assistant.
[Manifesto content ends]
```

This script generates a personalized manifesto using user input and saves it to a file.

CONCLUSION

Working on this project really shifted how I see open source. I used to treat Git like just another handy tool—useful, yeah, but nothing more. Now I get that it’s a whole way of thinking and collaborating, not just a piece of software. Now I get the bigger picture, the actual philosophy driving it.

Git wasn’t dreamed up in isolation—it was born out of everyday headaches that kept tripping developers up. It’s a tool built from real frustration to fix real problems and it shows the way it's designed, distributed and fast and surprisingly reliable, feels like something made by people who actually had skin in the game. Given all that history, it’s no shock Git is now stitched into nearly every software project. I used to breeze past the licensing fine print without a thought, but now I get why it matters. Then I dug in and realized it’s kind of the whole point. The GPL isn’t just lawyer speak it’s a promise that the code stays open for everyone, and that nobody gets to stroll across the bridge and yank it up behind them.

What also struck me was the community angle. Git is kept alive by developers scattered across the globe, people who've probably never met, working toward something shared. There's something genuinely impressive about that collaboration at a scale that most traditional organizations can't pull off.